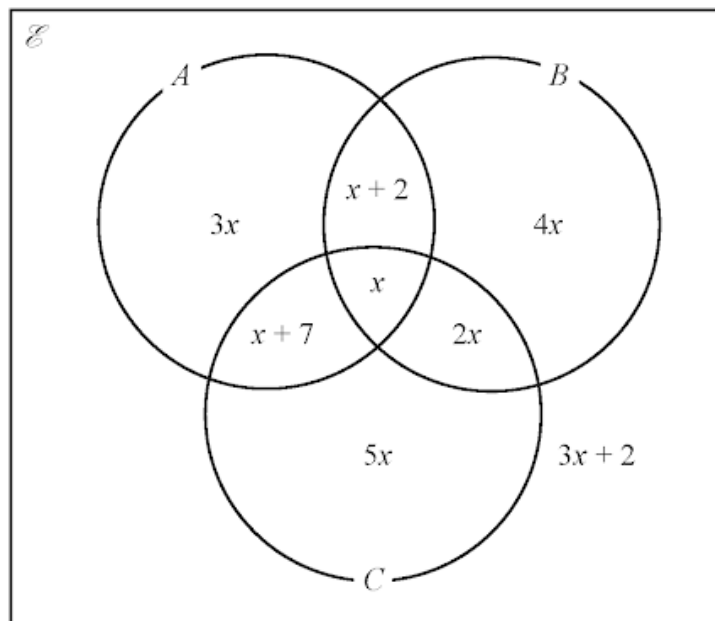


4MA1_2H_ch6_set_statistics_probability

Nguyen Khac Kiem PhD

1. 4MA1/2H_Winter_2024_Q21

The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

4 marks

2. 4MA1/2H_Winter_2024_Q13

The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

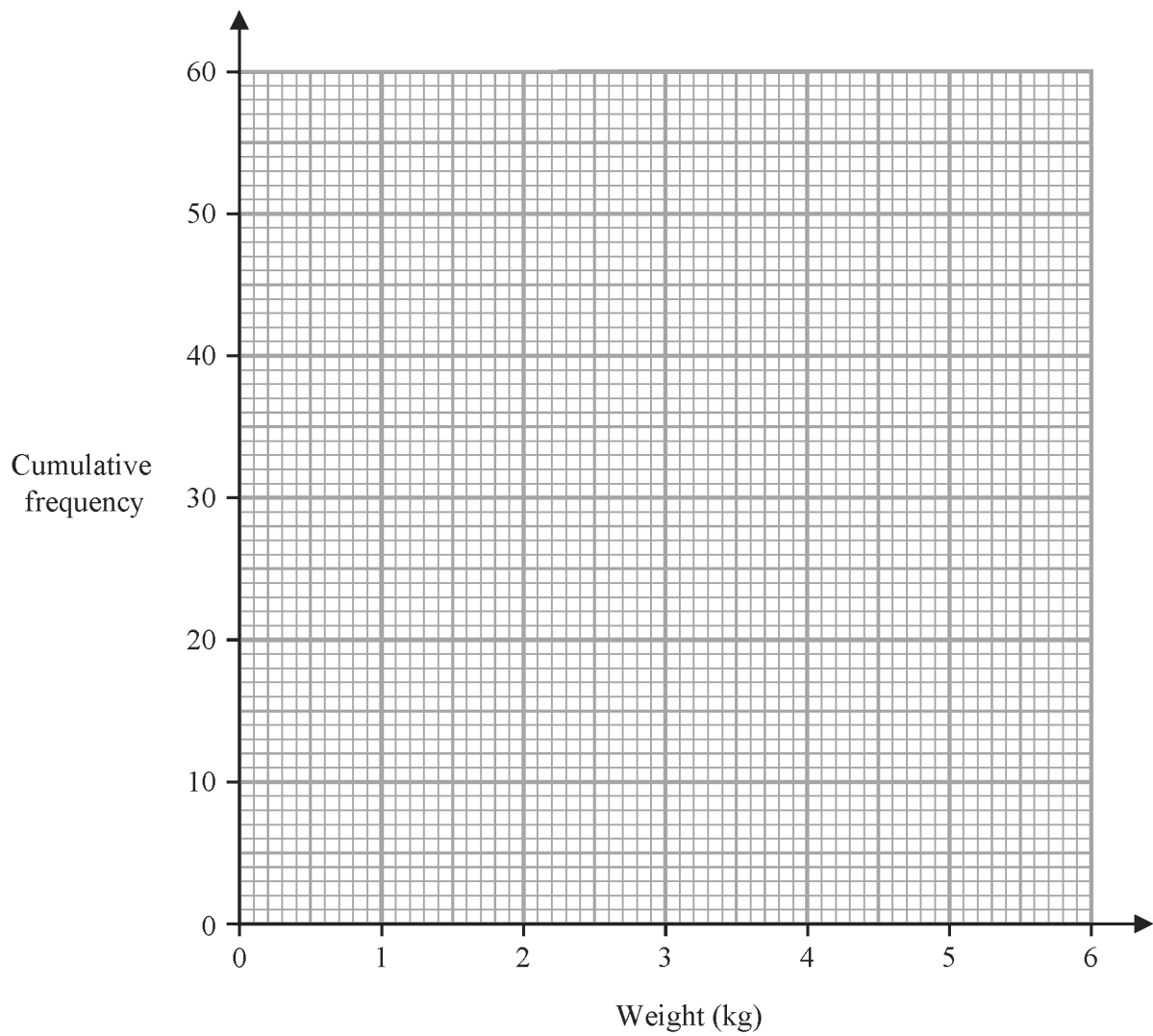
(a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the median weight of the 60 parcels.

..... kilograms
(1)

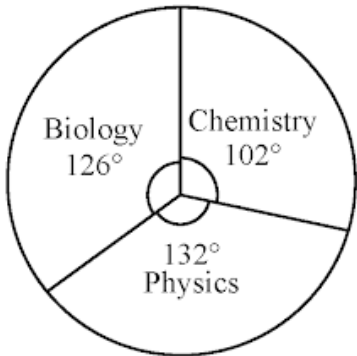
(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

.....
(2)

3. 4MA1/2H_Winter_2024_Q10

All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.
The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10
There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

5 marks

4. 4MA1/2H_Winter_2024_Q8

Kim puts 4 bananas in a bag.

The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.

The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

3 marks

5. 4MA1/2H_Winter_2024_Q5

Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

Number	1	2	3	4	5
Probability		0.14	0.17		0.21

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

4 marks

6. 4MA1/2HR_Summer_2024_Q14

Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8

The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

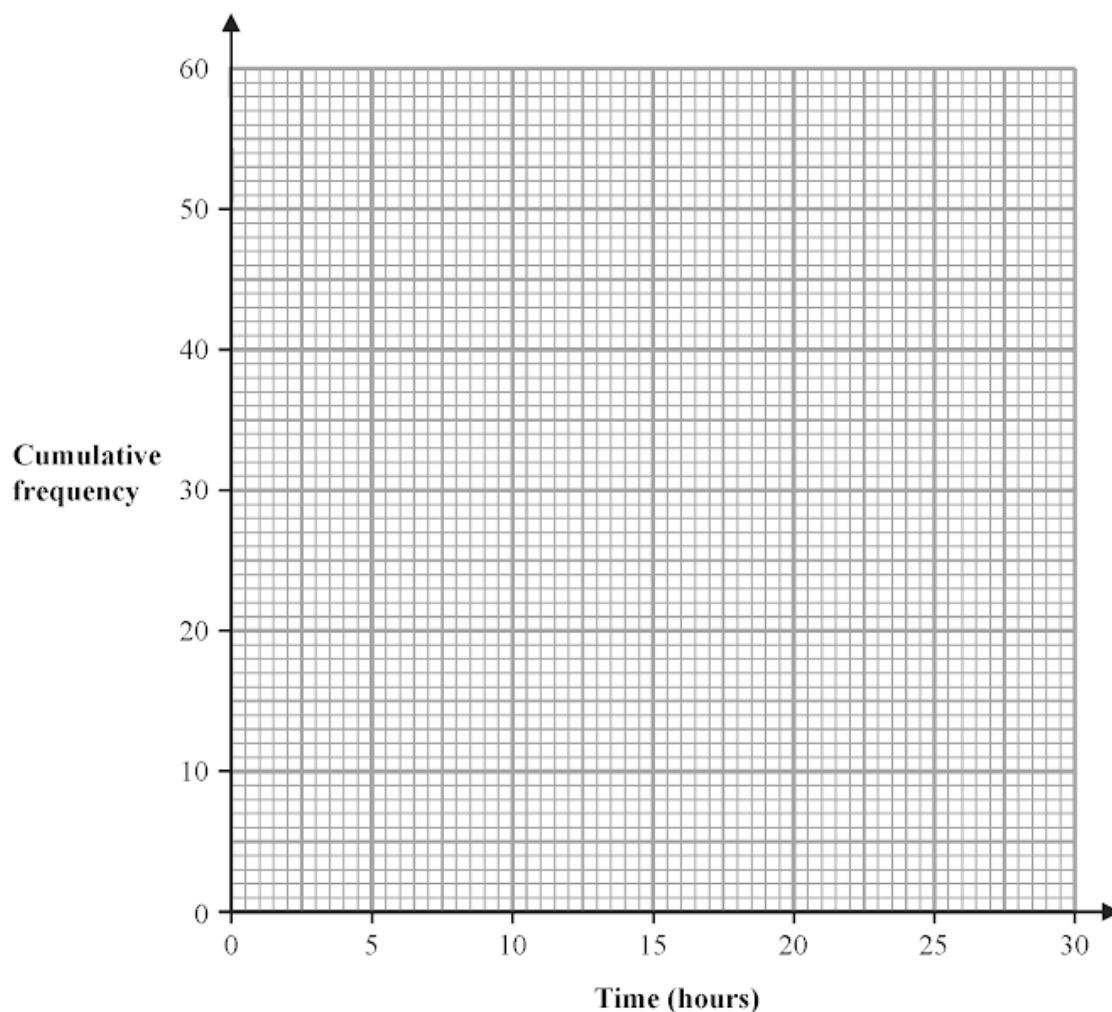
(3)

7. 4MA1/2HR_Summer_2024_Q11

The cumulative frequency table gives information about the time, in hours, that each of 60 workers spent working from home in one week.

Time (t hours)	Cumulative frequency
$0 < t \leq 5$	6
$0 < t \leq 10$	17
$0 < t \leq 15$	27
$0 < t \leq 20$	42
$0 < t \leq 25$	53
$0 < t \leq 30$	60

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(b) Use your graph to find an estimate for the interquartile range of the times.

..... hours
(2)

25 workers spent more than W hours working from home.

(c) Use your graph to find an estimate for the value of W

$W =$
(2)

One of the 60 workers is chosen at random.

This worker spent H hours working from home.

(d) Find the probability that $5 < H \leq 10$

(1)

8. 4MA1/2HR_Summer_2024_Q4

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{\text{factors of } 6\}$$

$$B = \{\text{prime numbers}\}$$

(a) List the members of the set

(i) $A \cup B$

(1)

(ii) A'

(1)

Harpreet states that $A \cap B = \emptyset$

Harpreet is incorrect.

(b) Explain why.

(1)

C is a set with 4 members such that

the set $A \cap C$ has 2 members

the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set C

(2)

9. 4MA1/2HR_Summer_2024_Q3

Here is a list of four numbers written in ascending order of size

x x y 15

where x and y are integers.

The numbers have

a median of 12.5

a range of 4

Find the value of x and the value of y

(2)

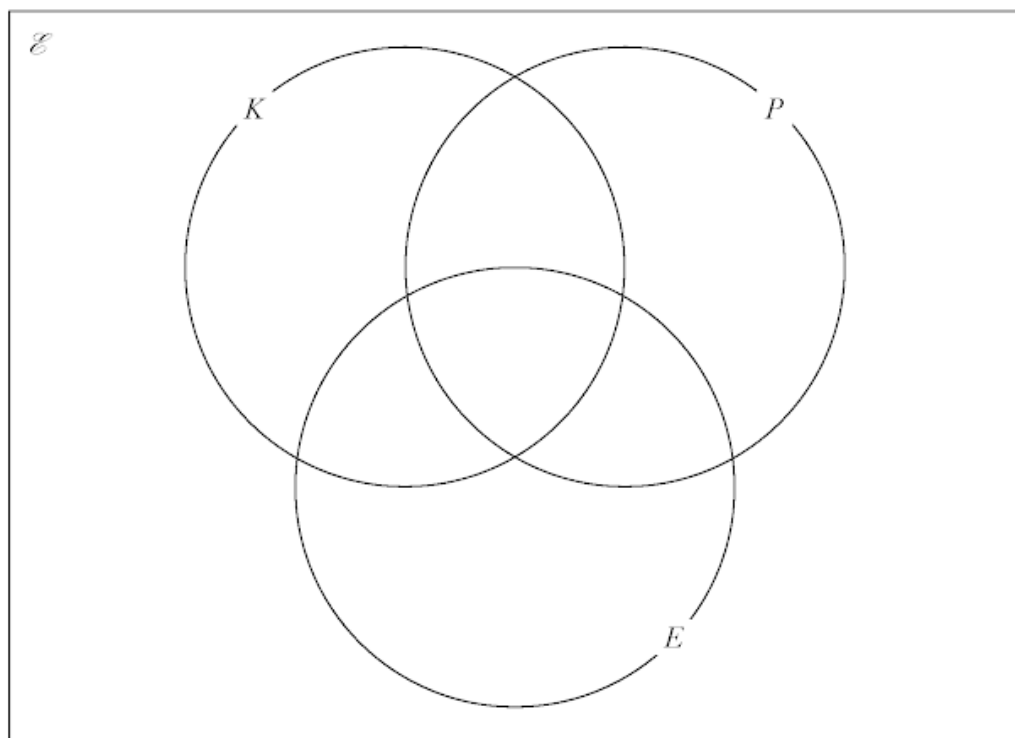
10. 4MA1/2H_Summer_2024_Q16

60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

One of the students is chosen at random.

Given that this student chose photography,

- (b) find the probability that this student also chose knitting.

(2)

- (c) Find $n(P \cap K')$

(1)

- (d) Find $n([P \cup E] \cap K)$

(1)

11. 4MA1/2H_Summer_2024_Q13

The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leq 10$	7
$10 < d \leq 20$	17
$20 < d \leq 30$	18
$30 < d \leq 40$	14
$40 < d \leq 50$	10
$50 < d \leq 60$	4

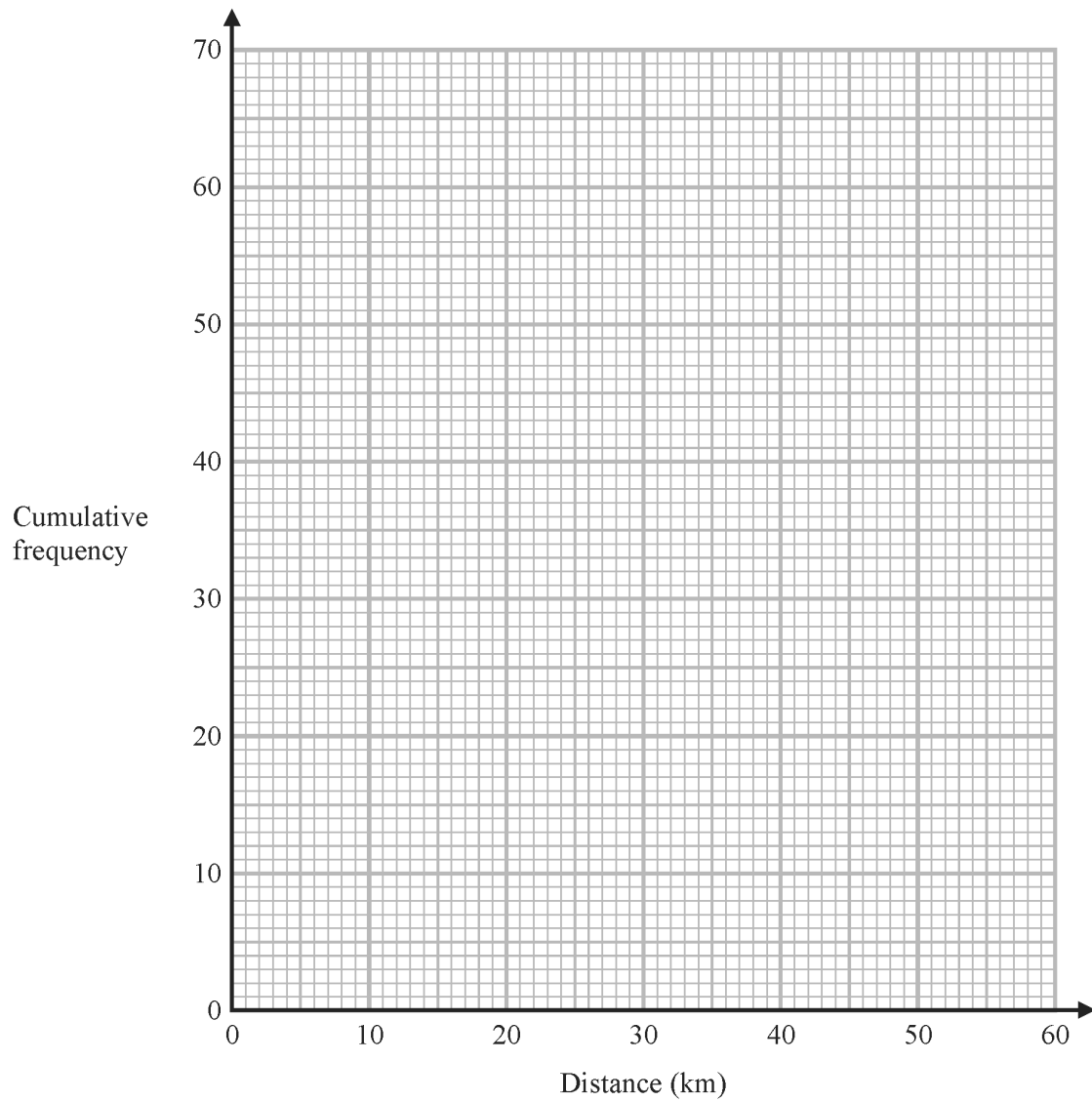
(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leq 10$	
$0 < d \leq 20$	
$0 < d \leq 30$	
$0 < d \leq 40$	
$0 < d \leq 50$	
$0 < d \leq 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

..... km
(2)

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

(2)

12. 4MA1/2H_Summer_2024_Q12

Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800

For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

(3)

13. 4MA1/2HR_Winter_2023_Q20

A , B and C are three sets.

$$n(A \cap B \cap C) = 5$$

$$n(A \cap B \cap C') = 2$$

$$n(A \cap C) = 5$$

$$n(A) = 17$$

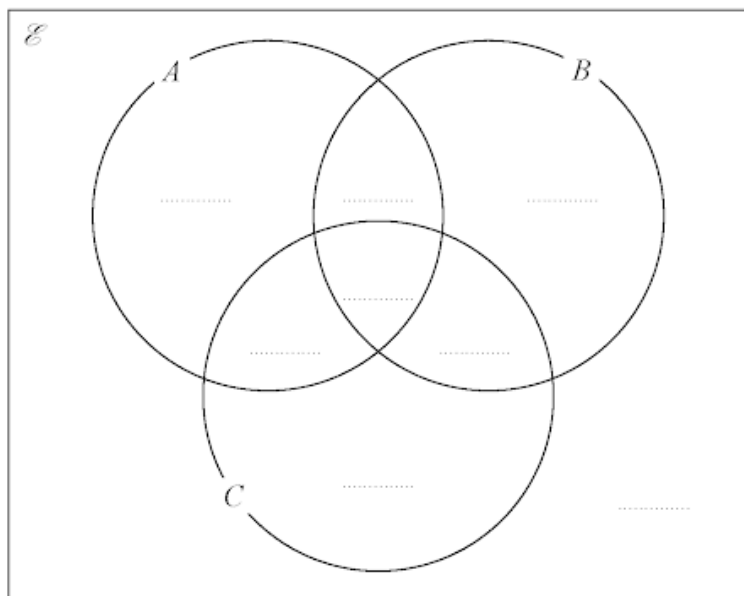
$$n([A \cup B \cup C]') = 3$$

$$n(A' \cap B \cap C') = 6$$

$$n(B \cap C) = 7$$

$$n(C) = 14$$

Complete the Venn diagram to show the number of elements in each region.



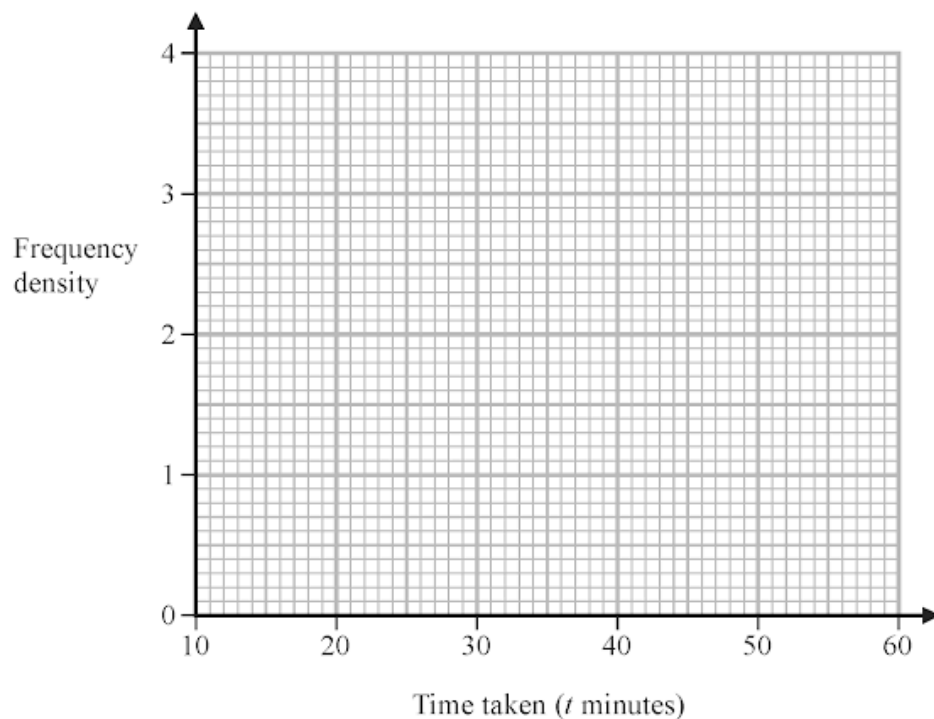
4 marks

14. 4MA1/2HR_Winter_2023_Q17

The table gives information about the time taken by each student in Year 11 to complete a homework task.

Time taken (t minutes)	Frequency
$10 < t \leq 25$	15
$25 < t \leq 30$	18
$30 < t \leq 50$	32
$50 < t \leq 60$	4

(a) On the grid, draw a histogram for this information.



(3)

One of these students who took 50 minutes or less and more than 25 minutes to complete this homework task is chosen at random.

(b) Find an estimate for the probability that this student took 45 minutes or less to complete this homework task.

(2)

15. 4MA1/2HR_Winter_2023_Q15

Here are the numbers of aces that Rutger served in each of 11 tennis matches.

1 1 2 4 6 8 8 9 11 12 15

- (a) Find the interquartile range of the numbers of aces.
Show your working clearly.

.....
(2)

Kim also plays in 11 tennis matches.

For Kim

the median number of aces is 11

the interquartile range of the numbers of aces is 5

- (b) State, giving a reason, whether Rutger or Kim

- (i) served more aces on average,

.....
.....
.....
(1)

- (ii) was more consistent with the number of aces served.

.....
.....
.....
(1)
.....
.....
.....
.....
.....

16. 4MA1/2HR_Winter_2023_Q6

In a bag, there are only red counters, blue counters, green counters and yellow counters.

The total number of counters in the bag is 80

In the bag

the number of red counters is $x + 7$

the number of blue counters is $x - 11$

the number of green counters is $3x$

Jude takes at random a counter from the bag.

The probability that he takes a red counter is $\frac{1}{4}$

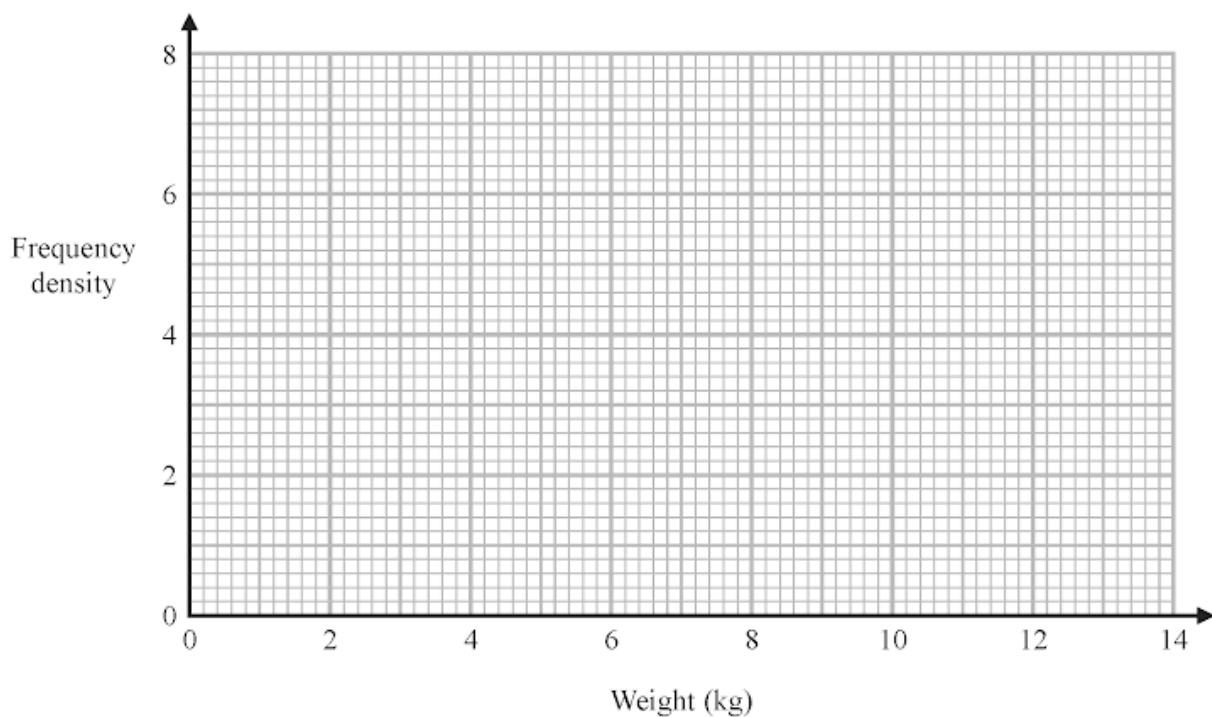
Work out the probability that Jude takes a yellow counter.

17. 4MA1/2H_Winter_2023_Q18

The table gives information about the weights, in kg, of the parcels that Pedro delivers on Monday.

Weight (w kg)	Frequency
$0 < w \leq 2$	12
$2 < w \leq 3$	7
$3 < w \leq 6$	15
$6 < w \leq 9$	12
$9 < w \leq 14$	9

(a) On the grid, draw a histogram for this information.



(3)

One of the parcels that Pedro delivered on Monday is chosen at random.

(b) Using the information in the table, find an estimate for the probability that this parcel weighs more than 7 kg.

(2)

18. 4MA1/2H_Winter_2023_Q15

Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

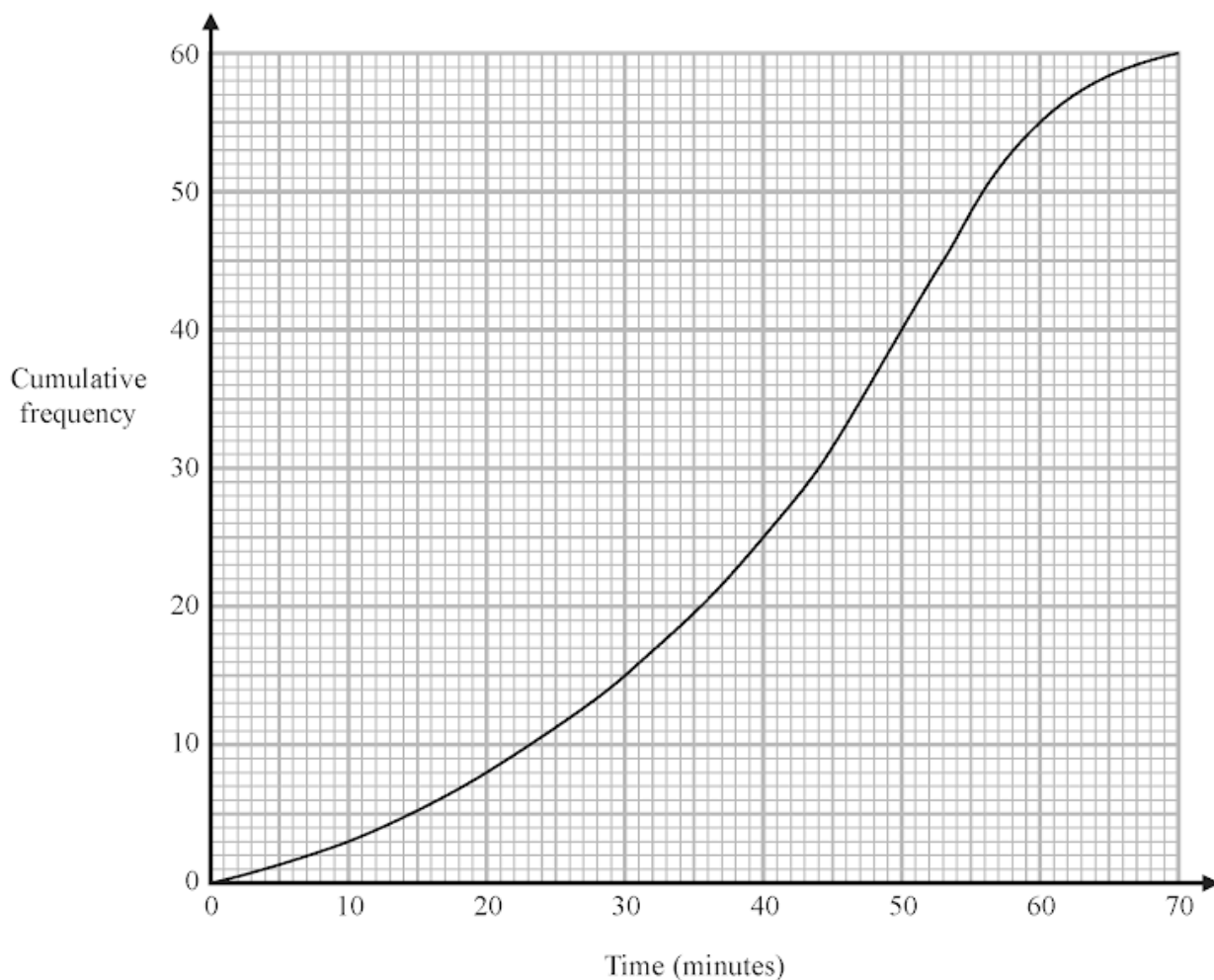
The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

3 marks

19. 4MA1/2H_Winter_2023_Q12

The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.



- (a) Use the graph to find an estimate for the median time people took to shop in the market.

..... minutes
(1)

- (b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

.....
(2)

(c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

Time taken to shop in the market (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	5
$20 < m \leq 30$	
$30 < m \leq 40$	
$40 < m \leq 50$	
$50 < m \leq 60$	
$60 < m \leq 70$	

20. 4MA1/2H_Winter_2023_Q3

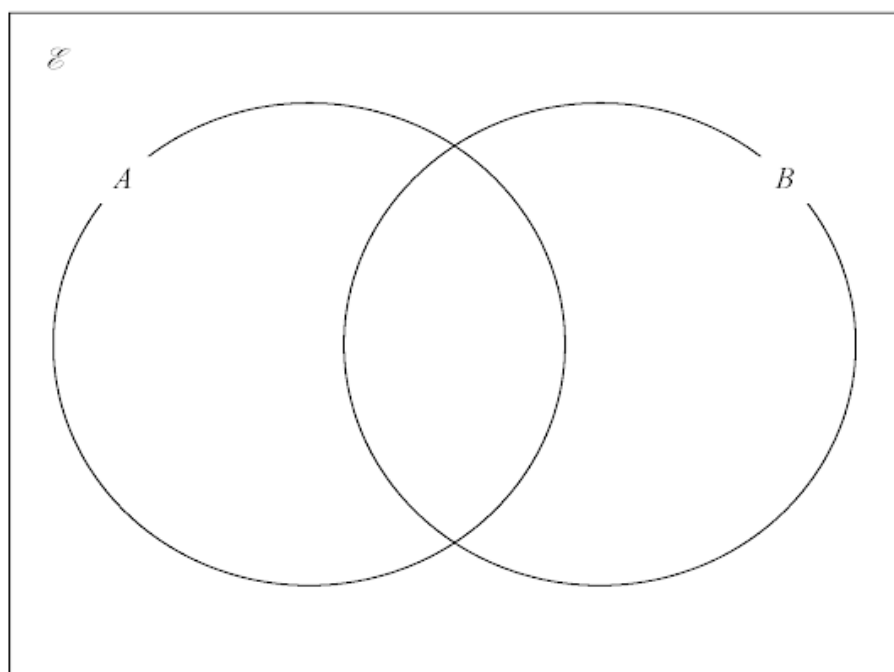
$$\mathcal{E} = \{11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

$$A = \{\text{even numbers}\}$$

$$A \cap B = \{12, 16, 20\}$$

$$(A \cup B)' = \{17, 19\}$$

Complete the Venn diagram for the sets $\mathcal{E}$, A and B



3 marks

21. 4MA1/2HR_Summer_2023_Q19

30 adults booked to stay in a hotel.

19 adults booked breakfast

15 adults booked dinner

4 adults did not book breakfast or dinner

Some adults booked breakfast **and** dinner.

Meihui chooses at random two of the 30 adults.

Work out the probability that these two adults each booked breakfast **and** dinner.

.....
4 marks

22. 4MA1/2HR_Summer_2023_Q15

Oswaldo has a biased coin.
He spins the coin three times.

The probability that the coin lands on a head three times is $\frac{27}{64}$

Work out the probability that the coin will land on a tail three times.

.....
3 marks

23. 4MA1/2HR_Summer_2023_Q14

Akari played a computer game eleven times.
Here are her scores.

25 20 28 27 26 22 23 29 20 29 26

(a) Find the interquartile range of her scores.

.....
(3)

Machi played the same computer game eleven times.

The interquartile range for Machi's scores was 9

(b) Who had the more consistent scores, Akari or Machi?

Give a reason for your answer.

.....
(1)

24. 4MA1/2H_Summer_2023_Q20

There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

.....

3 marks

25. 4MA1/2H_Summer_2023_Q13

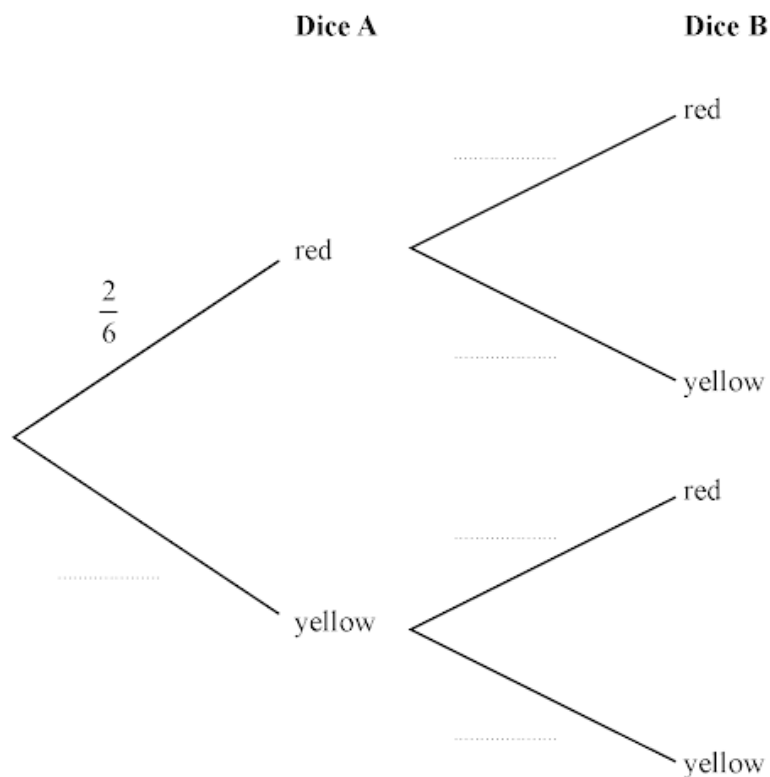
Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.

Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

(2)

26. 4MA1/2H_Summer_2023_Q2

A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	green	yellow	brown	pink
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

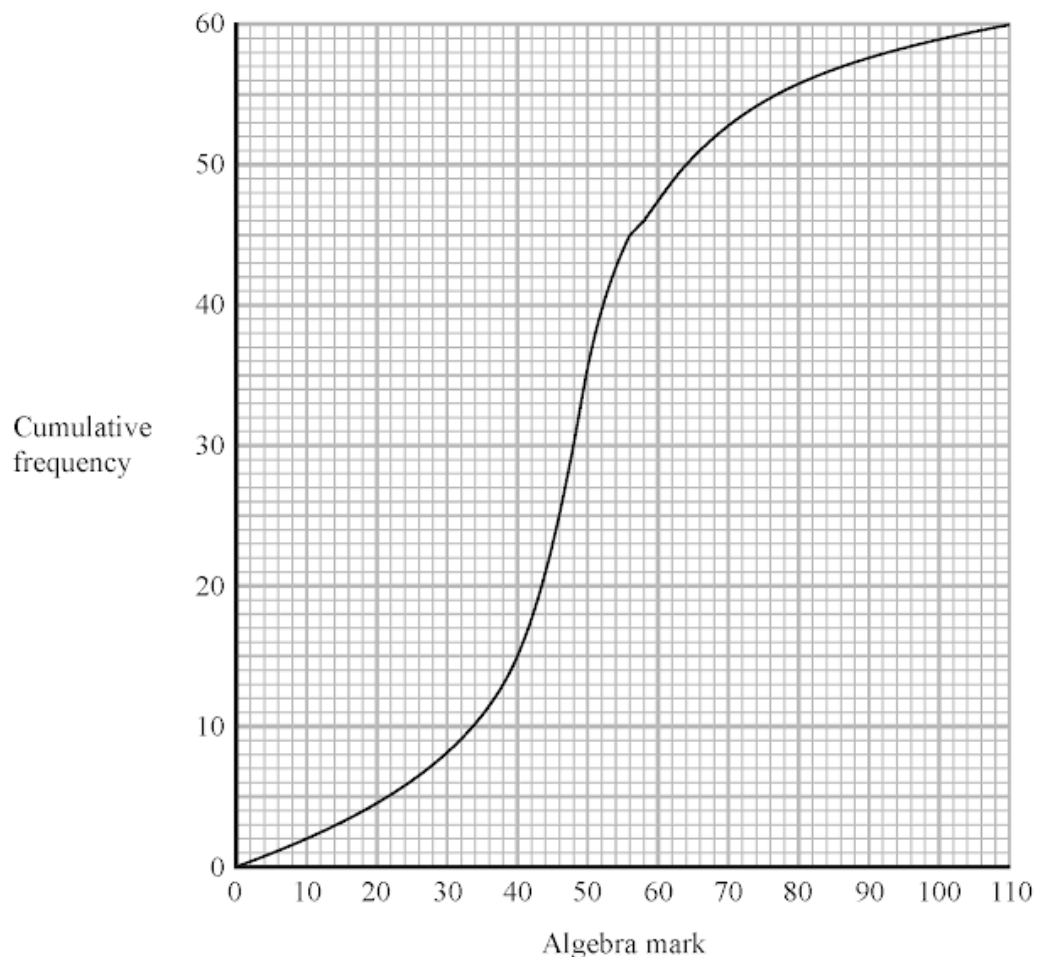
.....

3 marks

27. 4MA1/2HR_Winter_2022_Q14

A group of 60 students each sat an algebra test and a geometry test.
Each test was marked out of 110

The cumulative frequency graph gives information about the marks gained by the 60 students in the algebra test.



- (a) Use the graph to find an estimate for the median mark in the algebra test.

.....
(1)

- (b) Use the graph to find an estimate for the number of students who gained 58 marks or less in the algebra test.

.....
(1)

-
-
- (c) Use the graph to find an estimate for the interquartile range of the marks gained in the algebra test.

.....
(2)

The interquartile range of the marks gained in the geometry test is 9

Luis says

“The students’ marks are more spread out in the algebra test than in the geometry test.”

- (d) Is Luis correct?
Give a reason for your answer.

.....
(1)

To be awarded a grade A in the algebra test, a student had to gain a mark greater than 64

Two students are to be selected at random from the 60 students in the group.

- (e) Use the graph to find an estimate for the probability that both of these students were awarded a grade A in the algebra test.

.....
(3)

28. 4MA1/2HR_Winter_2022_Q1

A tin contains tea bags with a choice of four different flavours of tea.
The four flavours of tea are Assam or Darjeeling or Nilgiri or Rize.

Sara takes at random a tea bag from the tin.

The table shows each of the probabilities that the flavour of the tea Sara takes is Assam or Darjeeling or Rize.

Flavour of tea	Assam	Darjeeling	Nilgiri	Rize
Probability	0.38	0.24		0.16

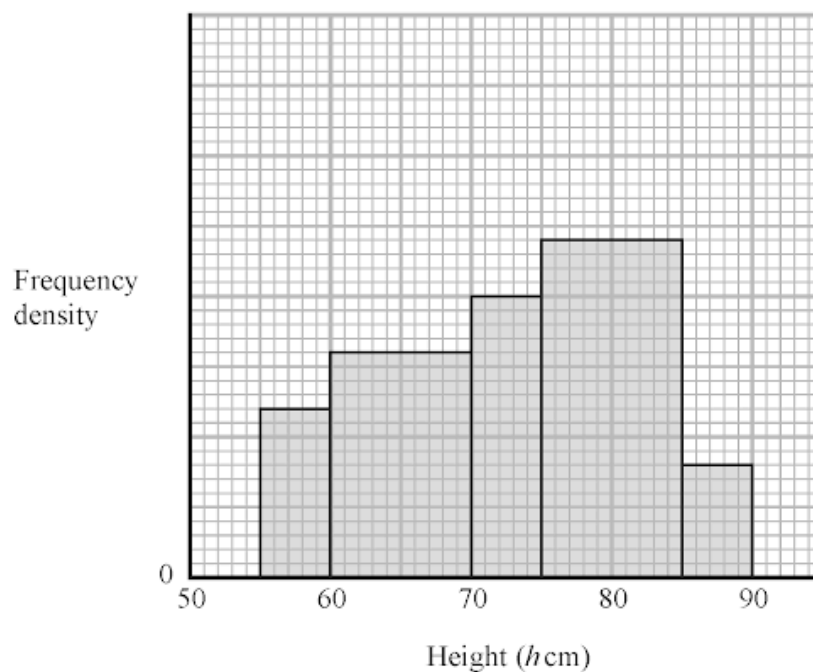
- (a) Work out the probability that the flavour of the tea Sara takes is Nilgiri.

.....
(2)

- (b) Work out the probability that the flavour of the tea Sara takes is either Darjeeling or Rize.

.....
(2)

The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

30. 4MA1/2H_Winter_2022_Q15

Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8 10 12 13 5 23 21 7 5 16 14

Find the interquartile range of his results.

..... km

3 marks
